

Assignment 6

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1. a.

$$\begin{vmatrix} 84 & 4 & 17 & 23 & 9 \\ 14 & 2 & 23 & 20 & 11 \\ 13 & 0 & 16 & 0 & 146 \\ 12 & 0 & 5 & 0 & 0 \\ 0 & 0 & 1/12 & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 84 & 4 & 23 & 9 \\ 14 & 2 & 0 & 11 \\ 13 & 0 & 0 & 146 \\ 12 & 0 & 0 & 0 \end{vmatrix}$$

- 12

$$= -12 \begin{vmatrix} 4 & 23 & 9 \\ 2 & 0 & 11 \\ 0 & 0 & 146 \end{vmatrix}$$

$$= \begin{vmatrix} 4 & 23 \\ 2 & 0 \end{vmatrix}$$

$$= (4) \cdot (0) - (23) \cdot (2) = -46$$

$$= \left(\frac{1}{12} \right) \cdot (-12) \cdot \left(\frac{1}{46} \right) \cdot (-46)$$

$$= 1.$$

b.

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 2 & 2 \\ 3 & 3 & 1 & 3 & 1 \\ 4 & 1 & 1 & 4 & 1 \\ 5 & 1 & 1 & 1 & 1 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & -1 & 0 & 0 \\ 3 & 0 & -2 & 0 & -2 \\ 4 & -3 & -3 & 0 & -3 \\ 5 & -4 & -4 & -4 & -4 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -1 & 0 & 0 \\ 0 & -2 & 0 & -2 \\ -3 & -3 & 0 & -3 \\ -4 & -4 & -4 & -4 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & -2 \\ -3 & 0 & -3 \\ -4 & -4 & -4 \end{vmatrix}$$

$$= -2 \begin{vmatrix} -3 & 0 \\ -4 & -4 \end{vmatrix}$$

$$= (-3) \cdot (-4) - (0) \cdot (-4) = 12$$

$$= (-2) \cdot (12)$$

$$= -24$$

$$2.a. \det [\bar{v}_1, \bar{v}_2 + 2\bar{v}_1, \bar{v}_3 + 3\bar{v}_1 + 4\bar{v}_2, \bar{v}_4 + 4\bar{v}_1 + 3\bar{v}_2 + 2\bar{v}_3]$$

Using column operation

$$C_2 \rightarrow C_2 - 2C_1, \quad C_3 \rightarrow C_3 - 3C_1, \quad C_4 \rightarrow C_4 - 4C_1$$

$$= \det [\bar{v}_1, \bar{v}_2, \bar{v}_3 + 4\bar{v}_2, \bar{v}_4 + 3\bar{v}_2 + 2\bar{v}_3]$$

$$C_3 \rightarrow C_3 - 4C_2, \quad C_4 \rightarrow C_4 - 3C_2$$

$$= \det [\bar{v}_1, \bar{v}_2, \bar{v}_3, \bar{v}_4 + 2\bar{v}_3]$$

$$= \det [\bar{v}_1, \bar{v}_2, \bar{v}_3, \bar{v}_4]$$

$$= \det A$$

$$= 2.$$

$$b. -1 \det [2\bar{v}_3, \bar{v}_1, 3\bar{v}_4 - \bar{v}_2]$$

$$= 6 \det [\bar{v}_3, \bar{v}_1, \bar{v}_4, \bar{v}_2]$$

$$= 6 \det [\bar{v}_1, \bar{v}_2, \bar{v}_3, \bar{v}_4]$$

$$= 6 \times \det A$$

$$= 6 \times 2$$

$$= 12$$

$$3. \det(A) = 0$$

$$\det \begin{bmatrix} a & b & c & d \\ e & 1-x & f & y \\ g & 0 & h & 0 \\ i & 0 & j & 0 \end{bmatrix} = 0$$

$$C_3 \rightarrow C_3 - \frac{h}{g} C_1$$

$$\det \begin{bmatrix} a & b & c - \frac{h}{g}a & d \\ e & 1-x & f - \frac{h}{g}e & y \\ g & 0 & 0 & 0 \\ i & 0 & j - \frac{h}{g}i & 0 \end{bmatrix} = 0$$

$$0 = 0 + \left(j + \frac{h}{g}i\right)(-1)^5 \det \begin{bmatrix} b & d \\ 1-x & y \end{bmatrix} + 0$$

$$0 = -\left(j - \frac{h}{g}i\right) \det \begin{bmatrix} b & d \\ 1-x & y \end{bmatrix}$$

$$0 = by - d(1-x)$$

$$0 \cdot b + 0 \cdot d = by - d(1-x)$$

$$\therefore y = 0 \quad \text{and} \quad 1-x = 0$$

$$x = 1$$